HACKATHON SUBMISSION · TIMETABL.AI

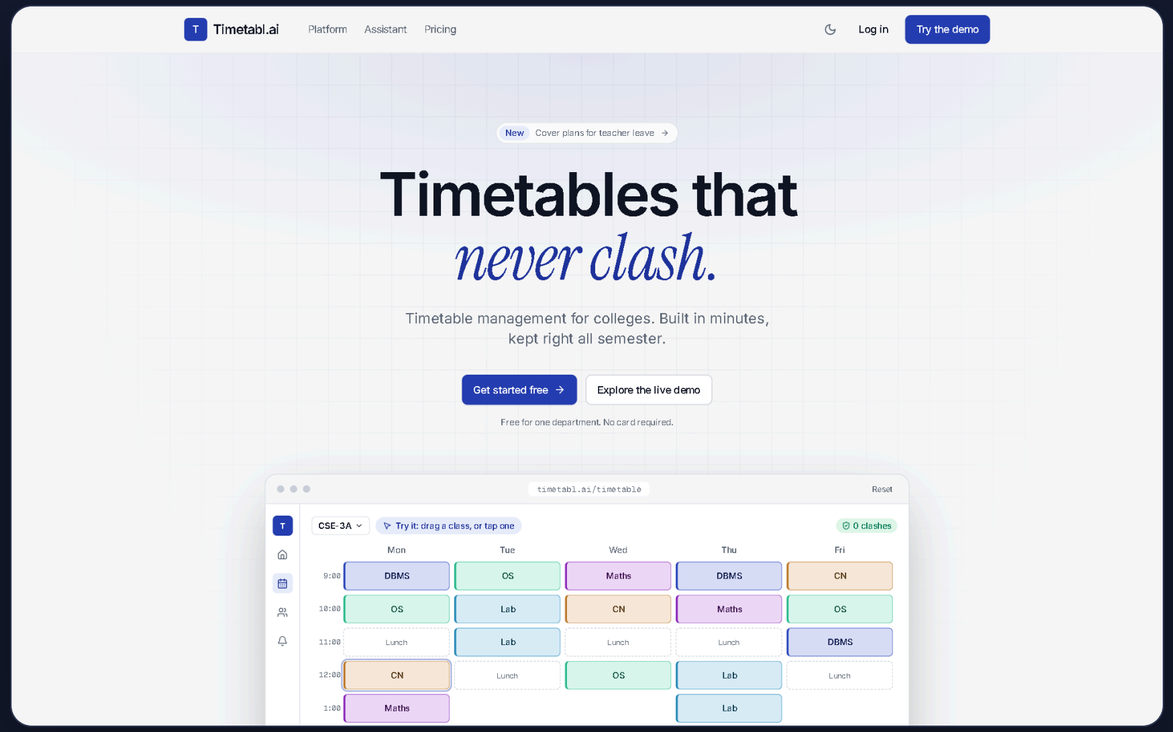
Timetables that *never clash.*

Clash-free college timetables in one click, and an AI assistant you can talk to that keeps them right all semester.

Live: timetabl-ai.vercel.app

Code: github.com/SamiyaS5904/Hackathon-PS_1-

Samiya Sehgal · URN 2302662 · GNDEC



PROBLEM STATEMENT

College timetables are made by hand, and they break every week

THE PROBLEM TODAY	WITH TIMETABL.AI
Two classes in one room, a teacher in two places	**0 clashes:** built by an algorithm, checked by a separate validator
Weeks of manual work for every new semester	**One Excel sheet** per department, **one click** for the whole college
A teacher is away and classes simply vanish	**A cover plan in seconds** with ranked substitutes
Every change means ten phone calls	**Just ask the assistant**, by voice or text; only those affected are told

HOW IT IS BUILT

Four parts, one rule: the algorithm decides, people apply

Web app

Next.js 15 · Vercel

Every role and screen, light and dark, phones.

AI assistant

Claude · OpenAI voice

Calls the backend with the user's own login.

Backend

FastAPI · OR-Tools · Fly.io

CP-SAT generator, independent validator, Excel/PDF.

Database

Supabase Postgres

Every row tagged by college.

Timetables come from the algorithm, never the AI

Every timetable and change passes the validator

The assistant prepares; a person clicks Apply

Each college sees only its own data

WHO USES IT

Many colleges, a clear role for everyone

Platform owner (Super Admin)
Plans, health, read-only or pause for every college. Never sees college data.

College A · College Admin
Setup, generate, publish; sees every department

College B · College Admin
Its own data, kept fully apart from College A

+ more colleges

CSE · Coordinator
Own department's data, approvals, leave cover

ECE · Coordinator
Own department's data, approvals, leave cover

ME · Coordinator
Own department's data, approvals, leave cover

+ more departments

Teachers
Own lectures, one-day moves, attendance, leave, office hours

Students
Own section and electives, free rooms, read-only assistant

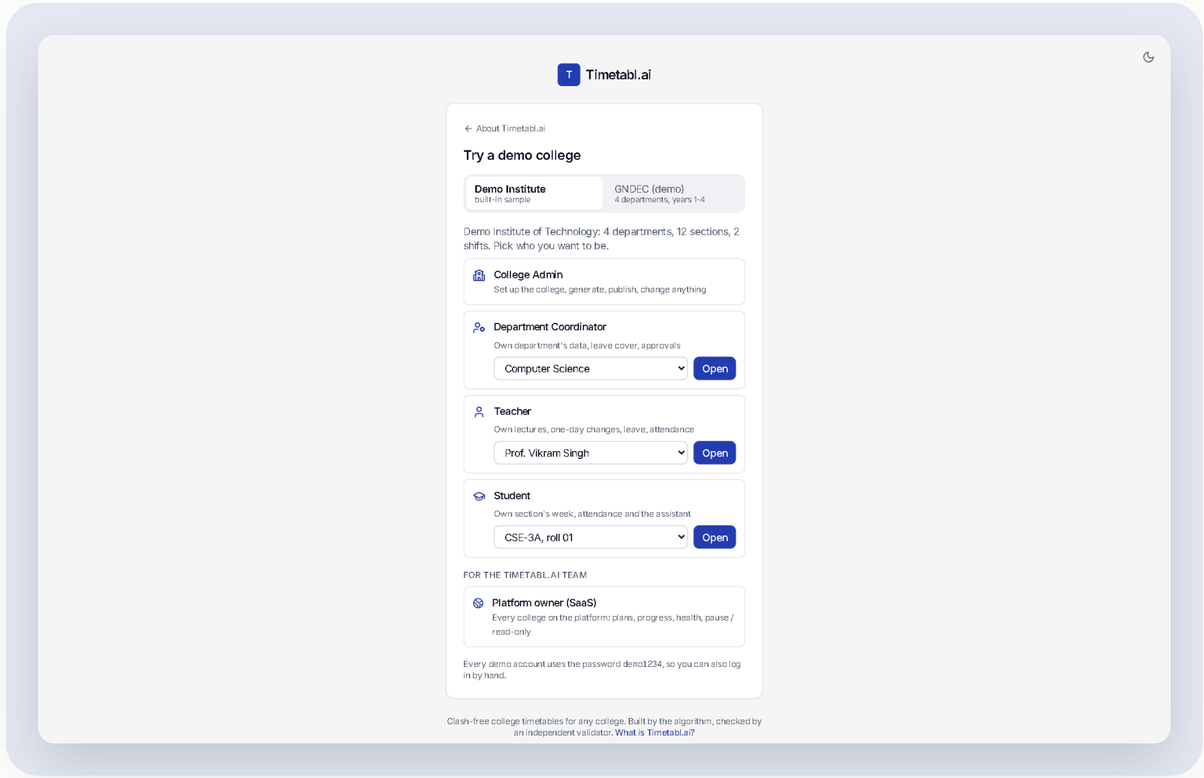
HOW TO TEST IT

Any role, one click, no sign-up

1. Open **timetabl-ai.vercel.app/demo**
2. Pick a college: Demo Institute or GNDEC (demo)
3. Pick a role → Open. Press **Ctrl+K** for the assistant

Role	Login (password demo1234)
College Admin	admin@demo.timetabl.ai
Coordinator	cse.coordinator@demo.timetabl.ai
Teacher	vikram.singh@demo.timetabl.ai
Student	cse3a.01@demo.timetabl.ai
Platform owner	owner@demo.timetabl.ai

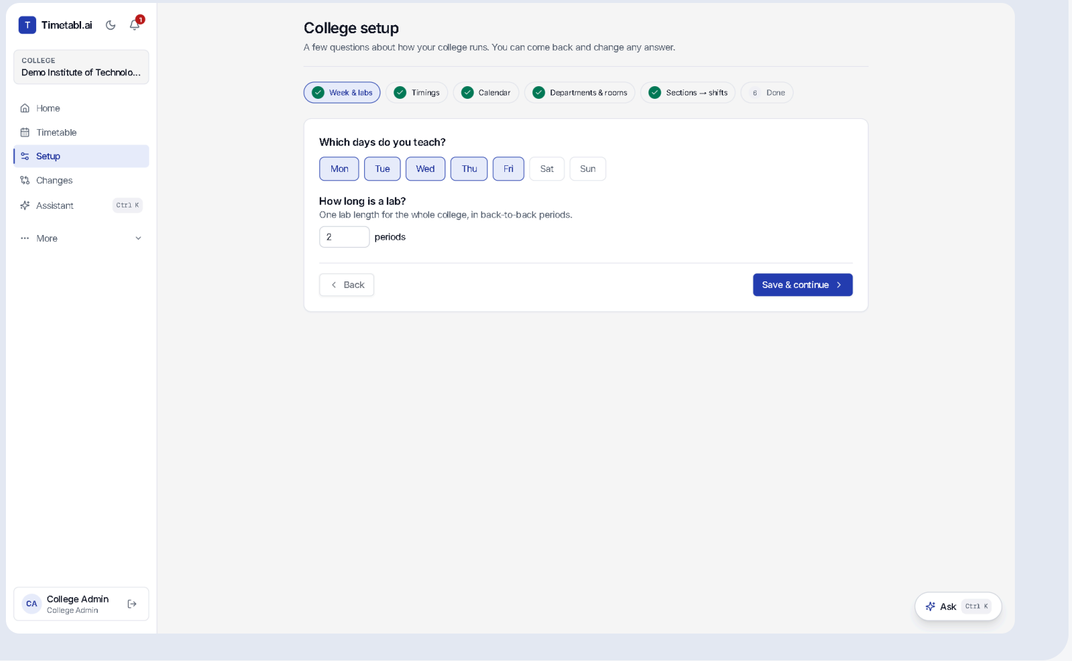
Testing dataset: docs/demo-data/GNDEC (5 department Excel files + a blank template)



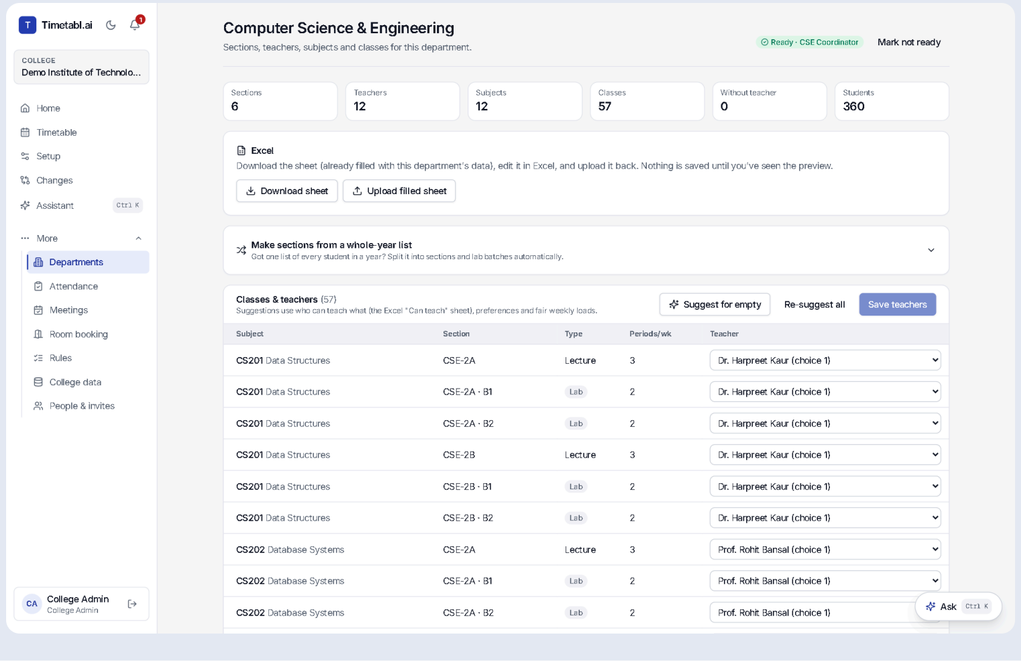
CORE · SET UP

Set up without help

> The Admin answers a short wizard; each Coordinator uploads one Excel sheet. Errors show per row, before anything is saved.



Setup: shifts, periods, mess window, holidays



Departments: Download sheet → Upload → Mark Ready

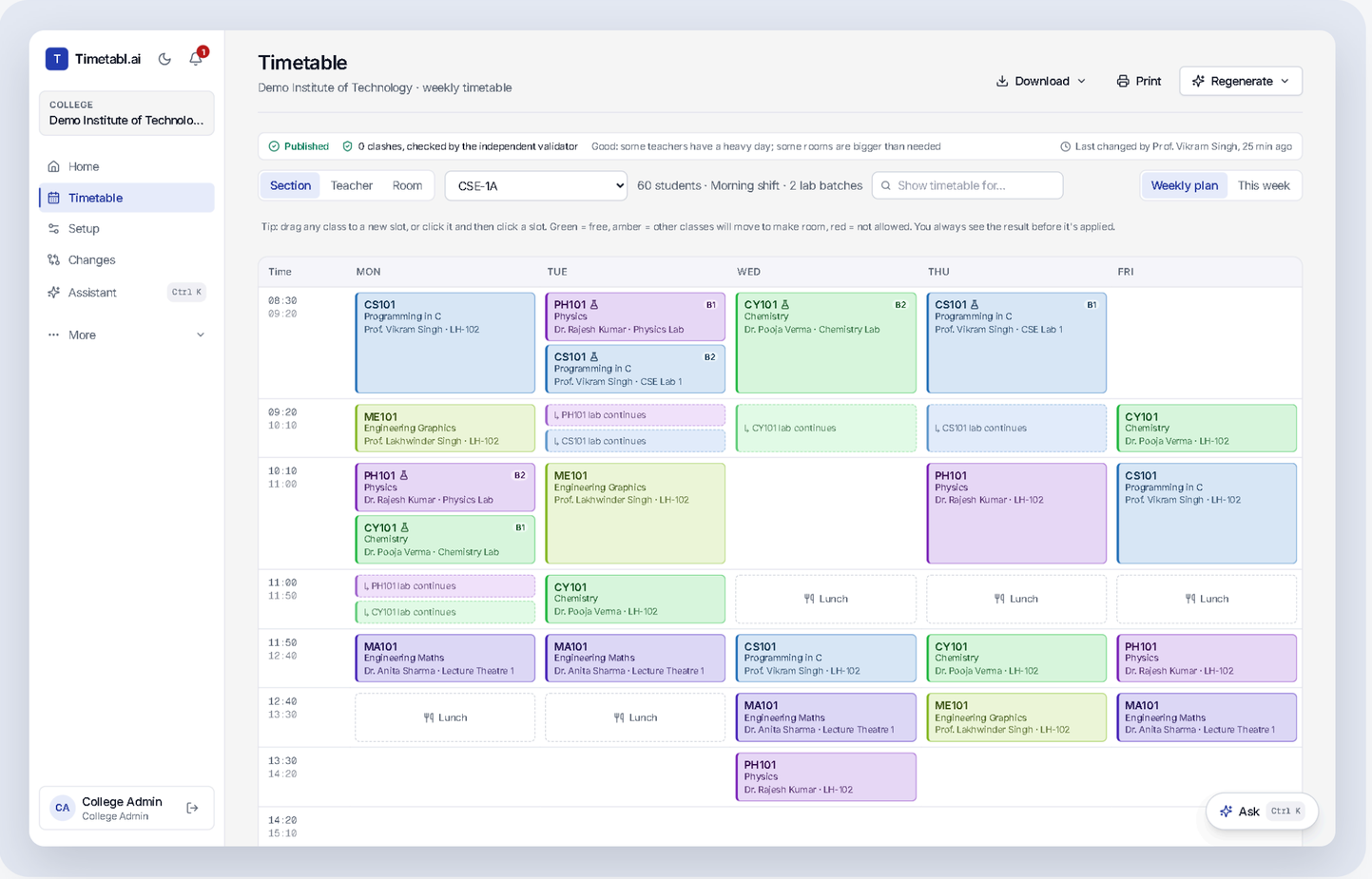
CORE · GENERATE

One click: the whole college

> **Every section, teacher and room gets a slot with 0 clashes, checked by an independent validator.**

Labs as blocks, combined classes, electives, 1 or 2 shifts, shared rooms, Must / Try rules.

1. Home → Generate
2. Check: Published · 0 clashes
3. Publish to everyone

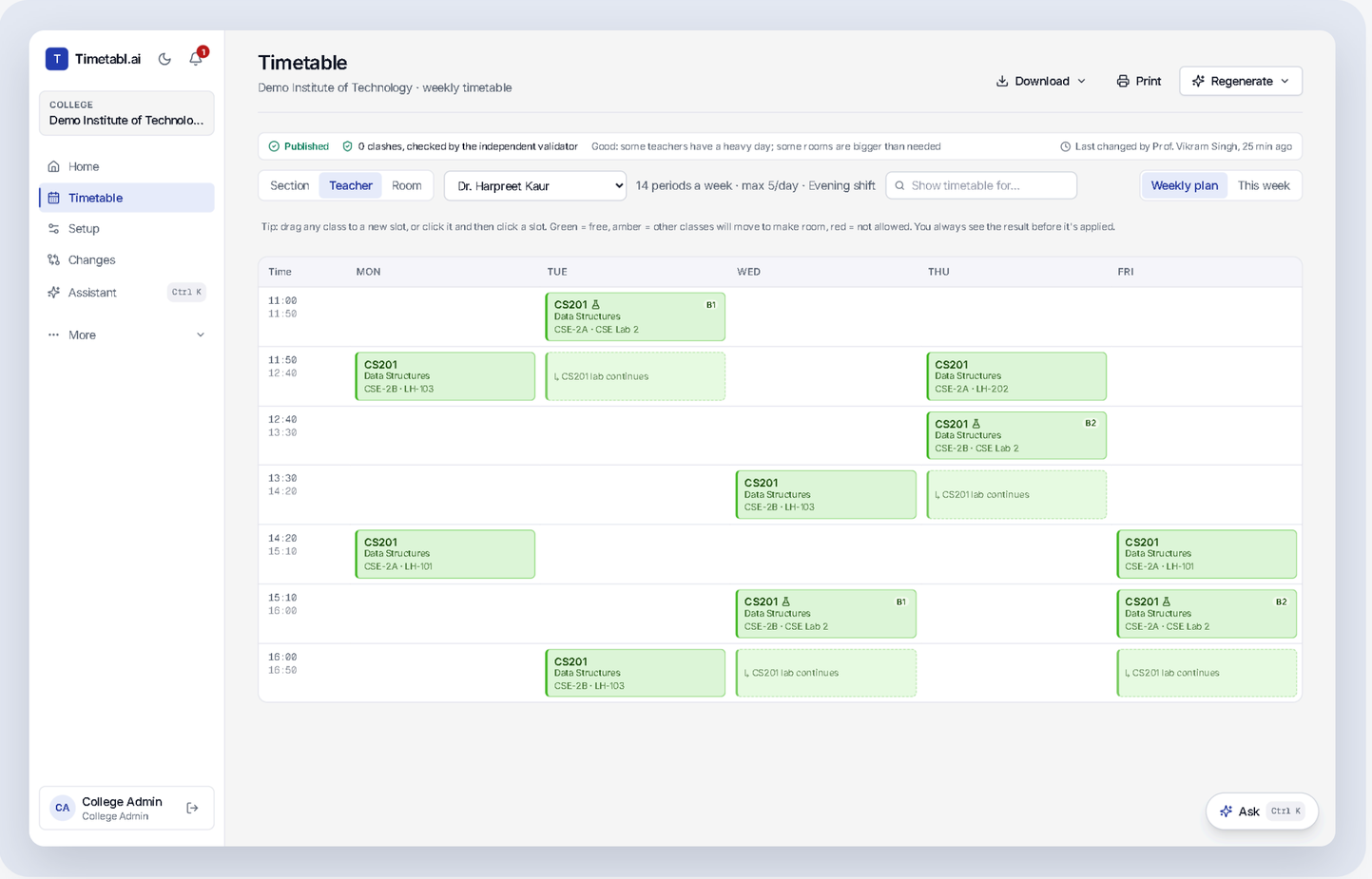


CORE · VIEW AND SHARE

Section, teacher and room views

> **Everyone sees only what is theirs, on any screen, and it prints straight to the notice board.**

1. Timetable → Section, Teacher or Room
2. Weekly plan, or This week with dates
3. Download → Excel or PDF, one or all

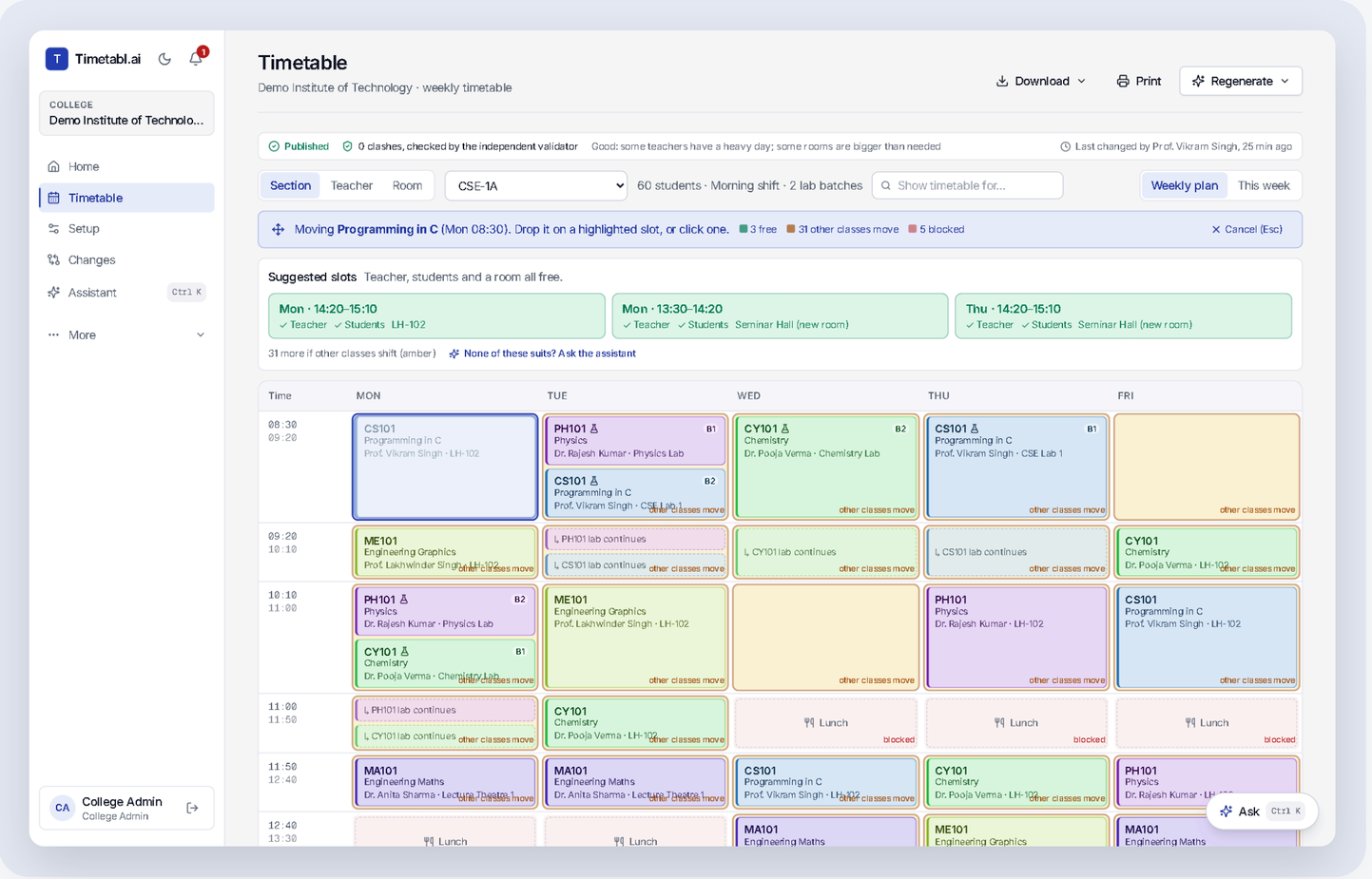


CORE · CHANGE SAFELY

Move a class: the app suggests where

Suggested slots already have the teacher, every student and a room free. You see the result before it applies.

1. Weekly plan → click or drag a class
2. Pick a suggested slot or a green cell
3. Review → Apply. None suits? Ask the assistant

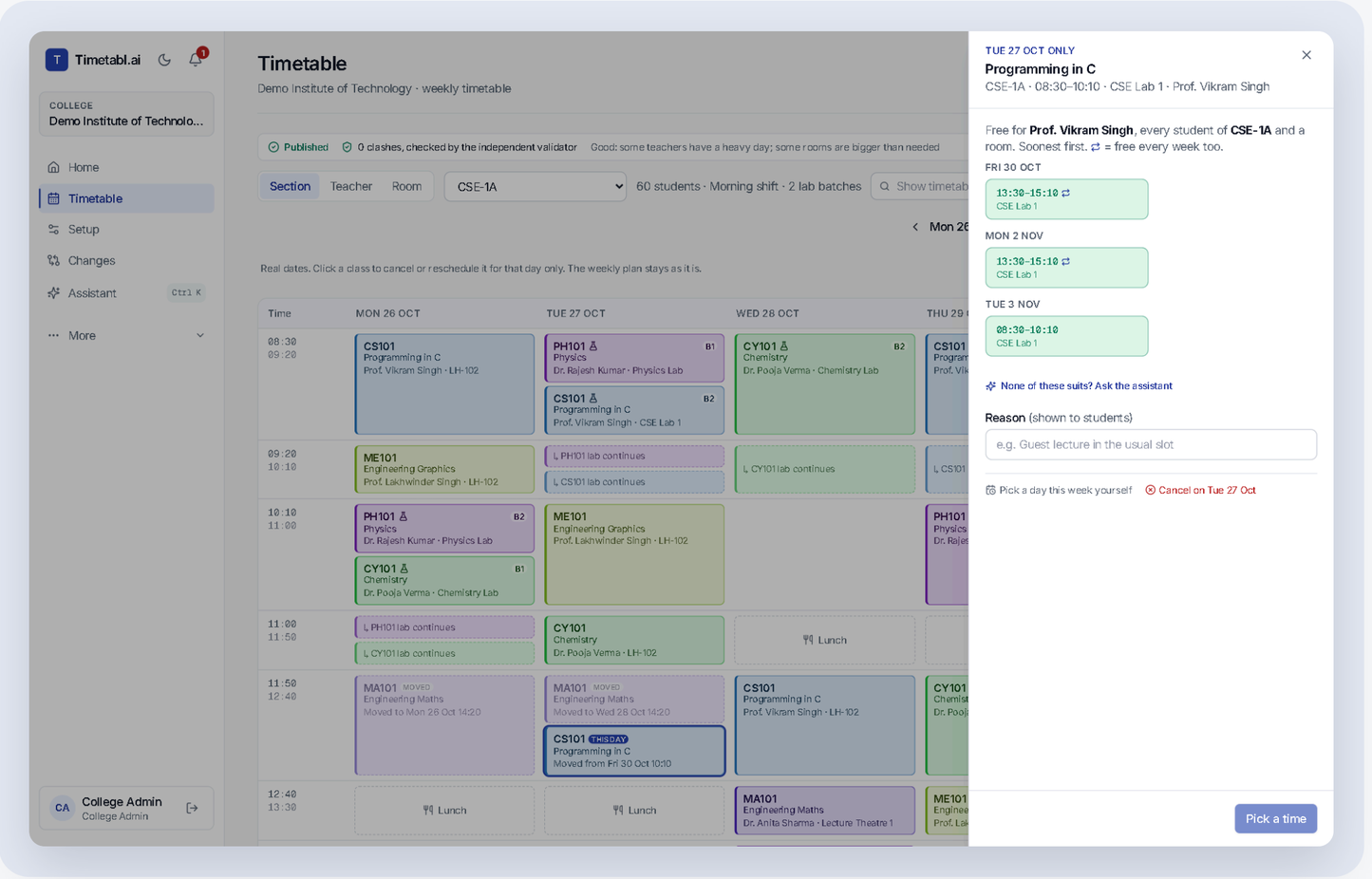


CORE · CHANGE SAFELY

One-day changes and makeup classes

Change one real date only; the weekly plan stays. The finder lists times when everyone is free.

1. Timetable → This week → click a class
2. Pick a free time, or cancel for the day
3. Students and teachers are told at once

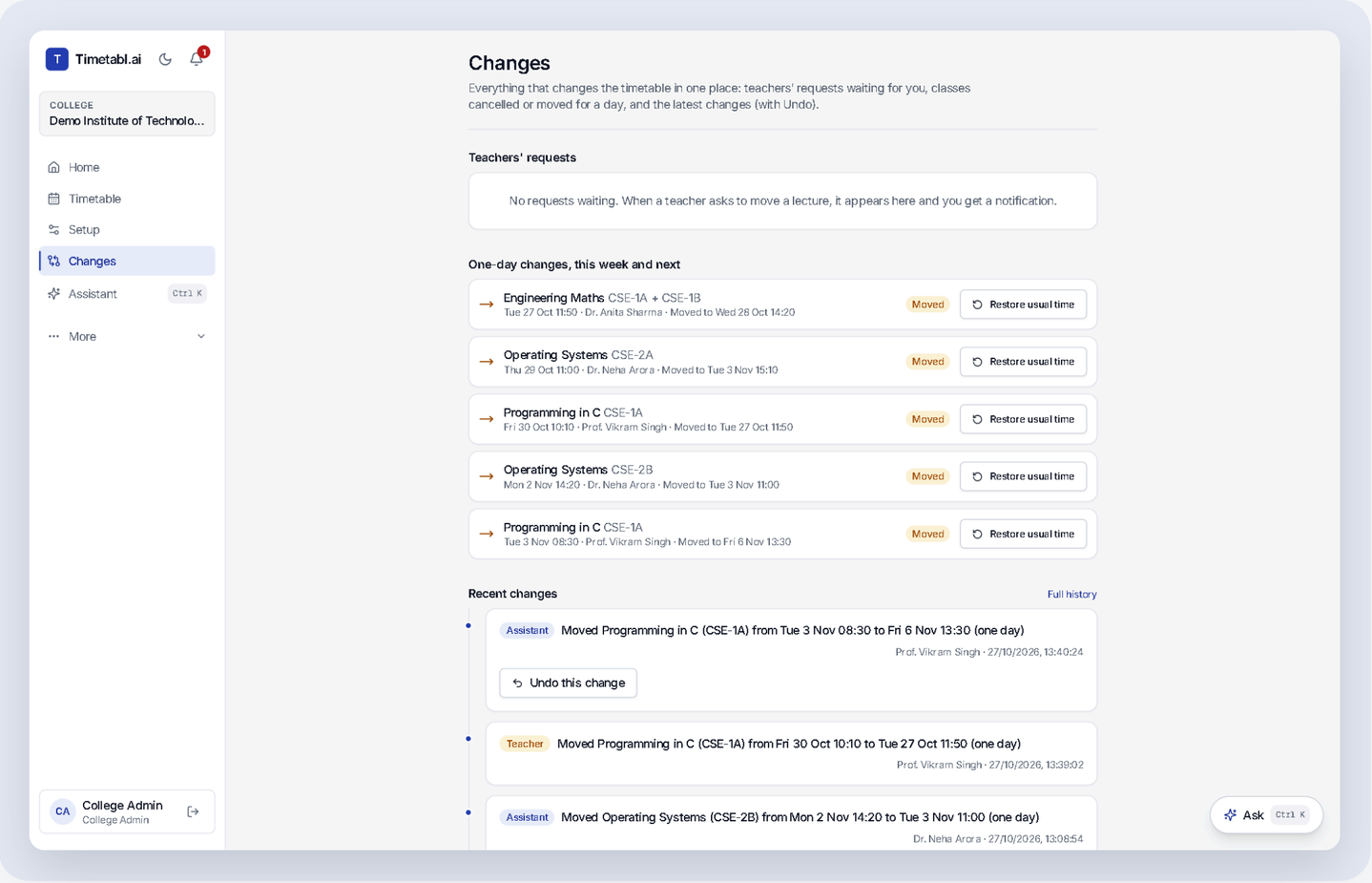


CORE · TRUST

Every change logged and undoable

Who, what, when and how: manual, drag, assistant or system. Only the people affected are notified.

1. Changes → this week's changes
2. Undo, or Restore usual time
3. Approve teachers' move requests

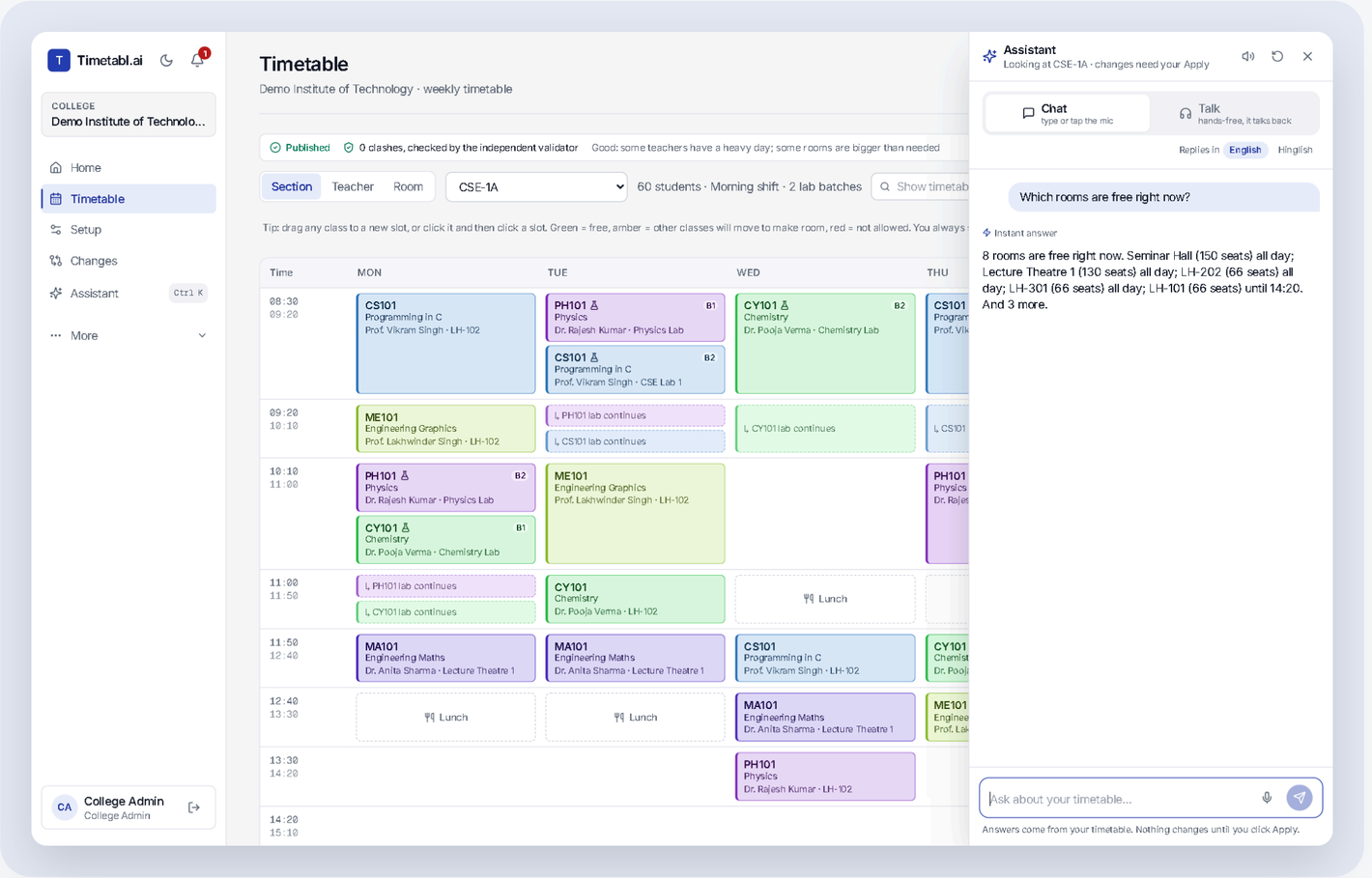


EXTRA · TALKING AI ASSISTANT

Talk to it. It does the change beside you.

Speak in English or Hinglish. It answers aloud, opens the right timetable next to the chat, and prepares the change. You just Apply.

1. Ctrl+K → Chat (type) or Talk (hands-free)
2. Say: “Move CSE-3A DBMS to Friday”
3. Watch it on screen → Apply

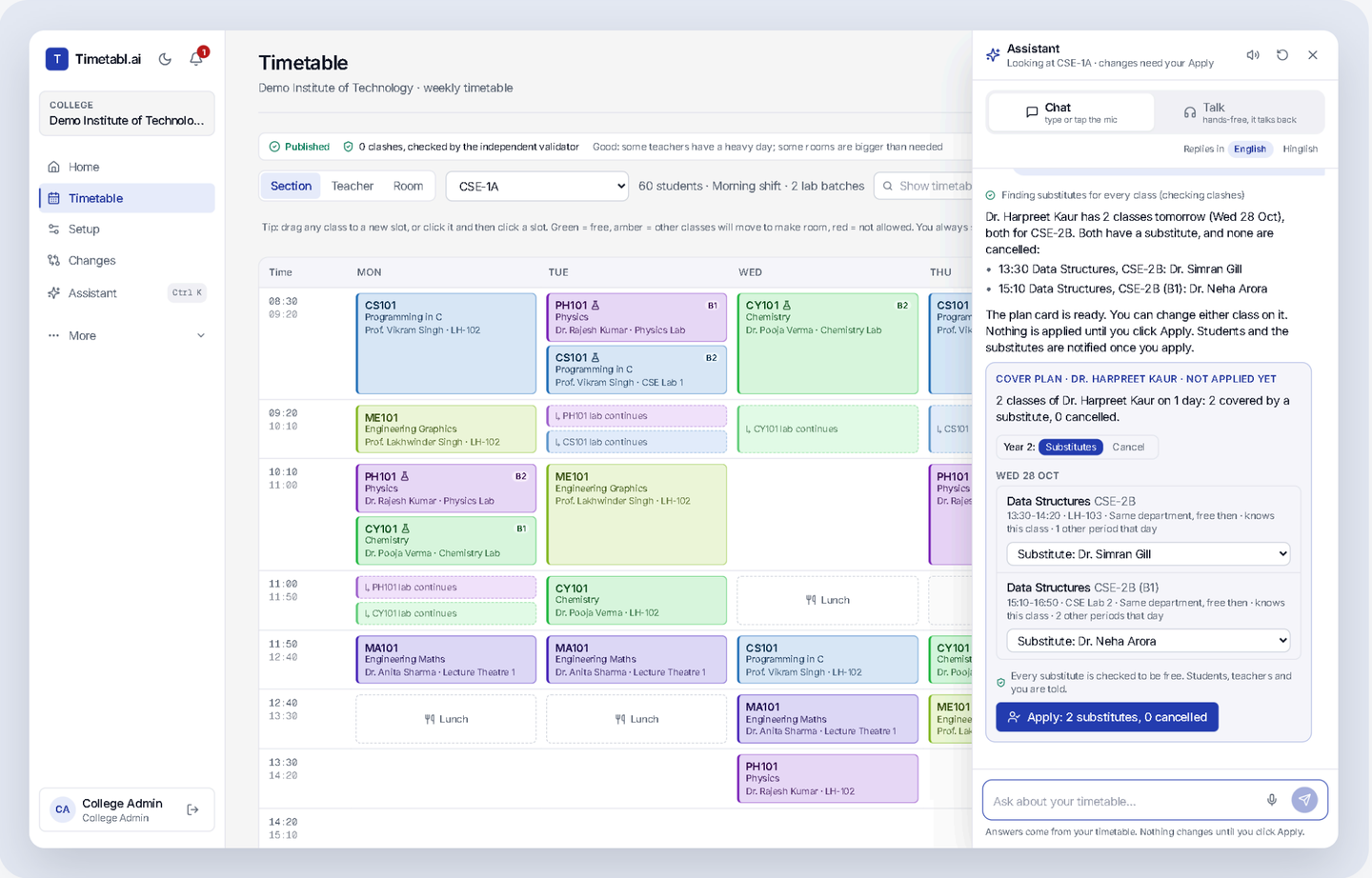


EXTRA · AGENTIC

Teacher away? A cover plan in seconds

Every affected class gets a ranked substitute (same subject, free then, fair load) or is cancelled.

1. Say: “Dr. Kaur is away tomorrow”
2. Check the plan card, change any pick
3. Apply → everyone involved is notified

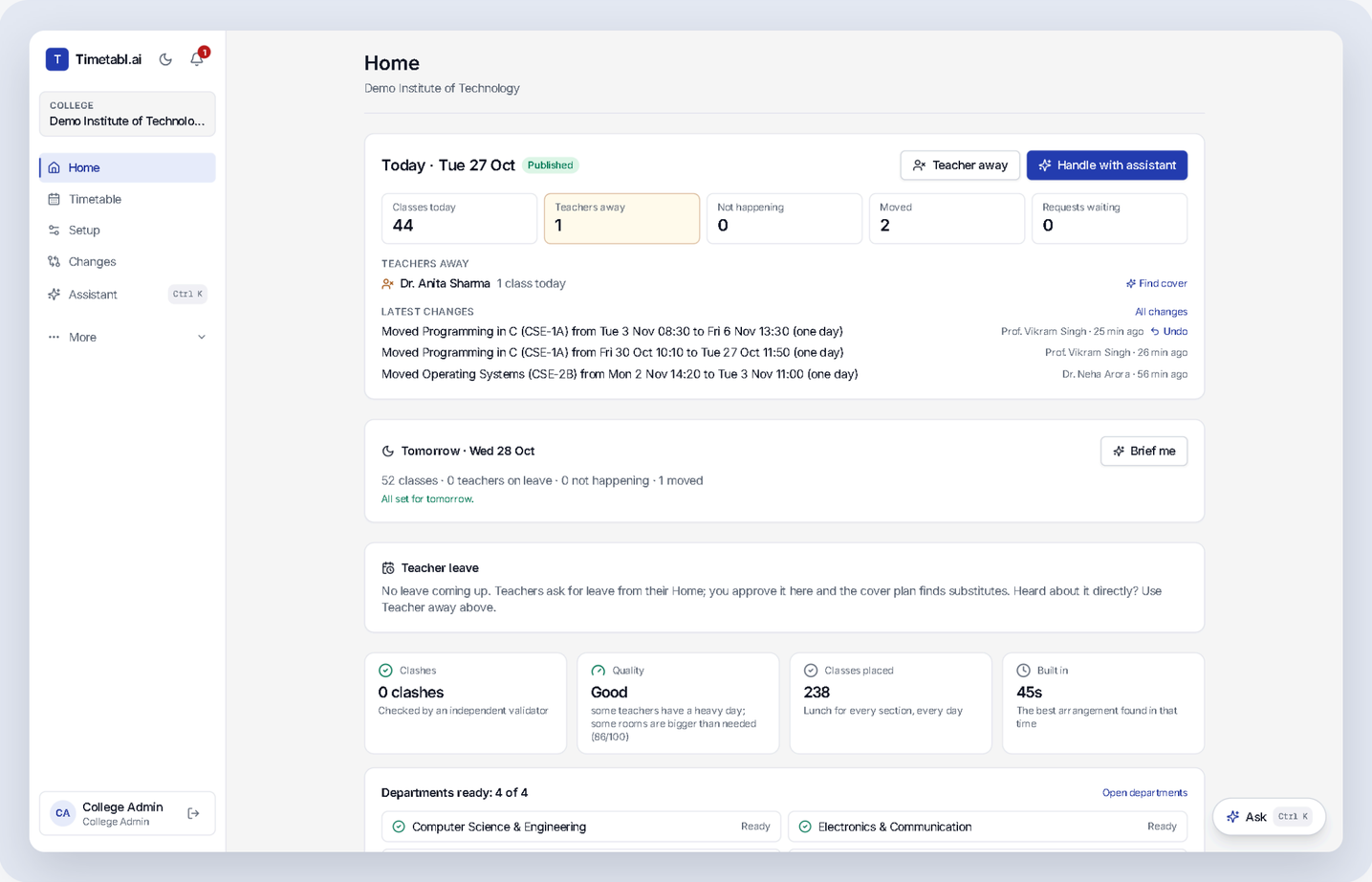


EXTRA · ADMIN AND COORDINATOR

Today at a glance

Teachers away, cancelled and moved classes, requests and tomorrow's brief, each one click from a fix.

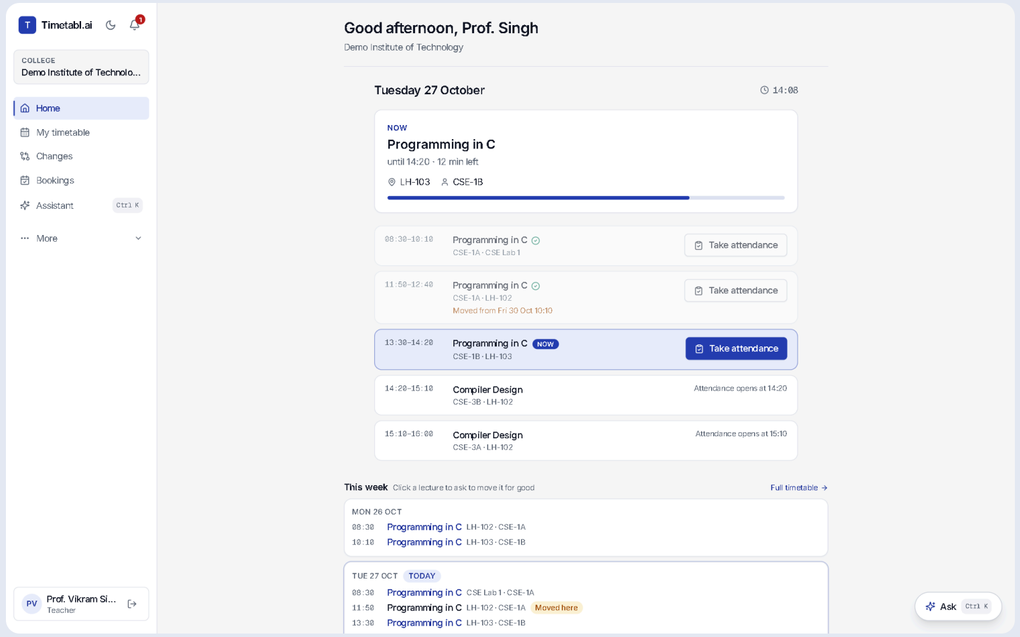
1. Log in → Home
2. Teachers away → Find cover
3. Brief me: tomorrow's gaps, early



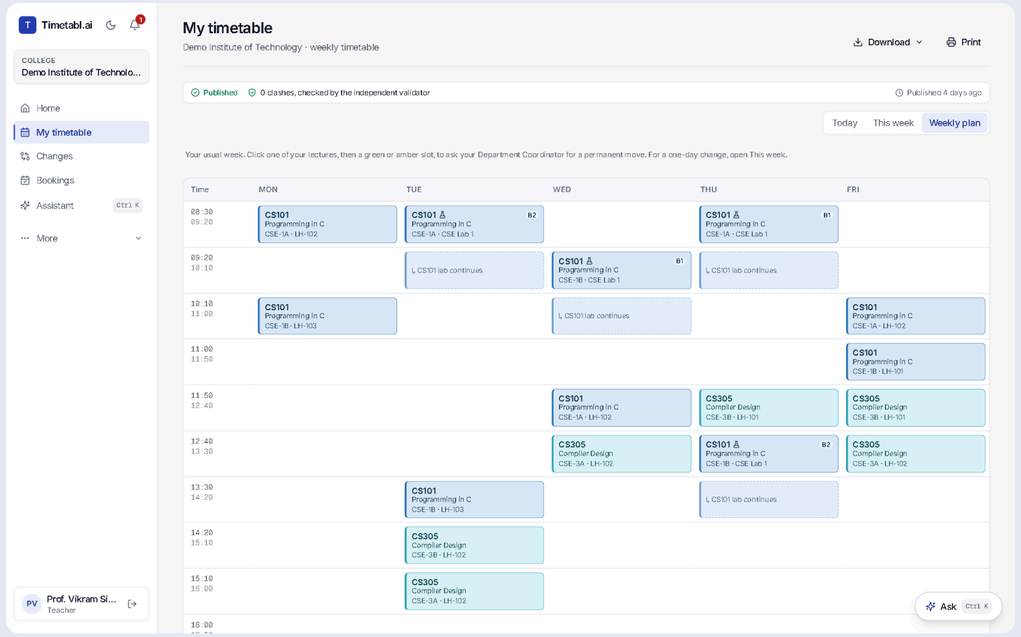
EXTRA · TEACHERS

Their day, and only their lectures

> **Now and next with a countdown, Take attendance in one tap. One-day moves are instant; permanent ones go to the Coordinator.**



Home: today, this week, Take attendance



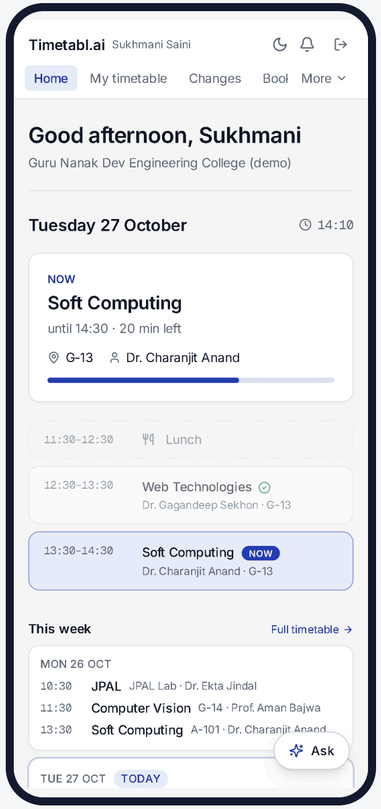
My timetable: click a lecture → ask to move it

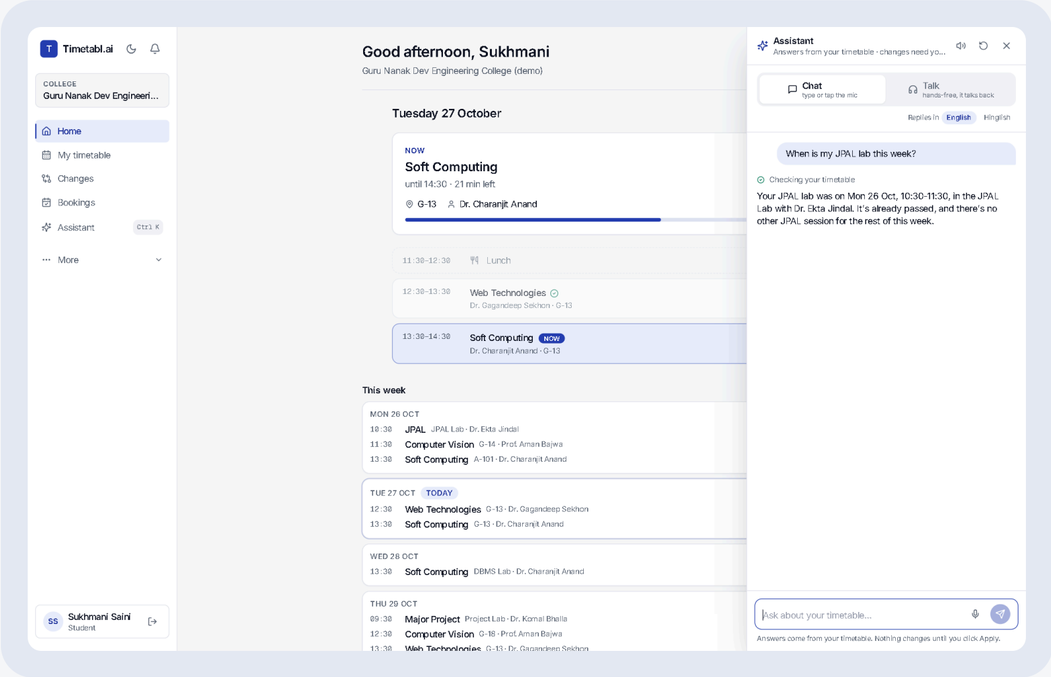
EXTRA · STUDENTS

Their week, on their phone

> **Today, this week, free rooms to sit in, attendance, and an assistant that answers (read-only).**

1. Demo → GNDEC → JPAL student
2. Ask: “When is my JPAL lab?”
3. Ask: “Where can I sit now?”

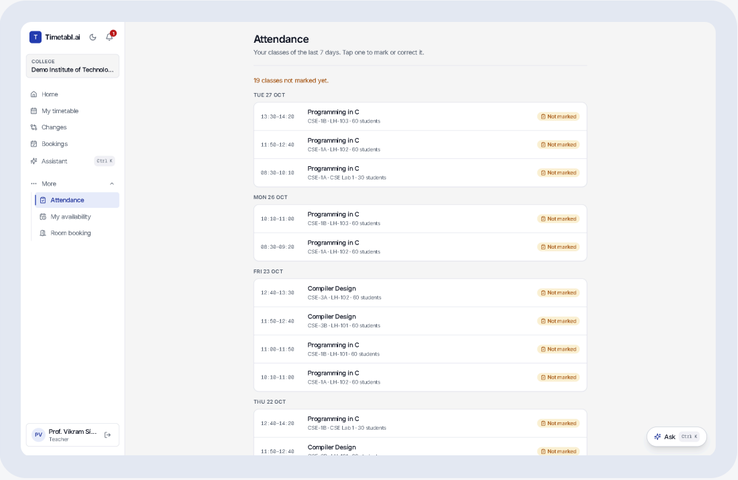




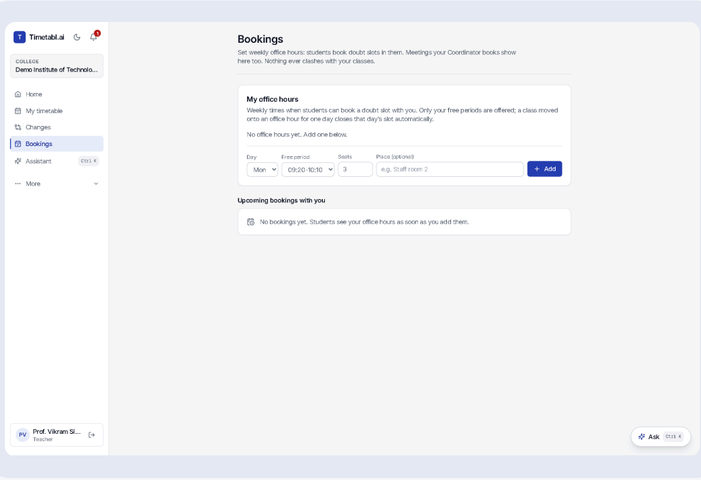
EXTRA · DAILY COLLEGE LIFE

Attendance, bookings and room events

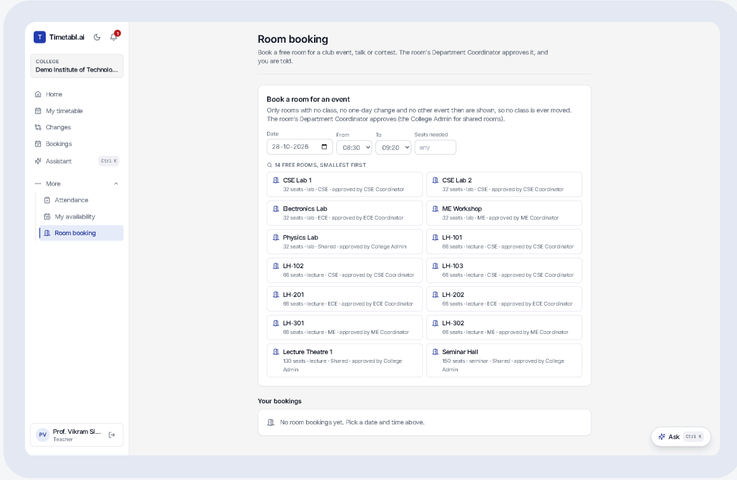
> **Built on the same timetable, so nothing booked ever clashes with a class.**



Attendance: all present by default; students see “can miss N”



Bookings: office hours → students book doubt slots



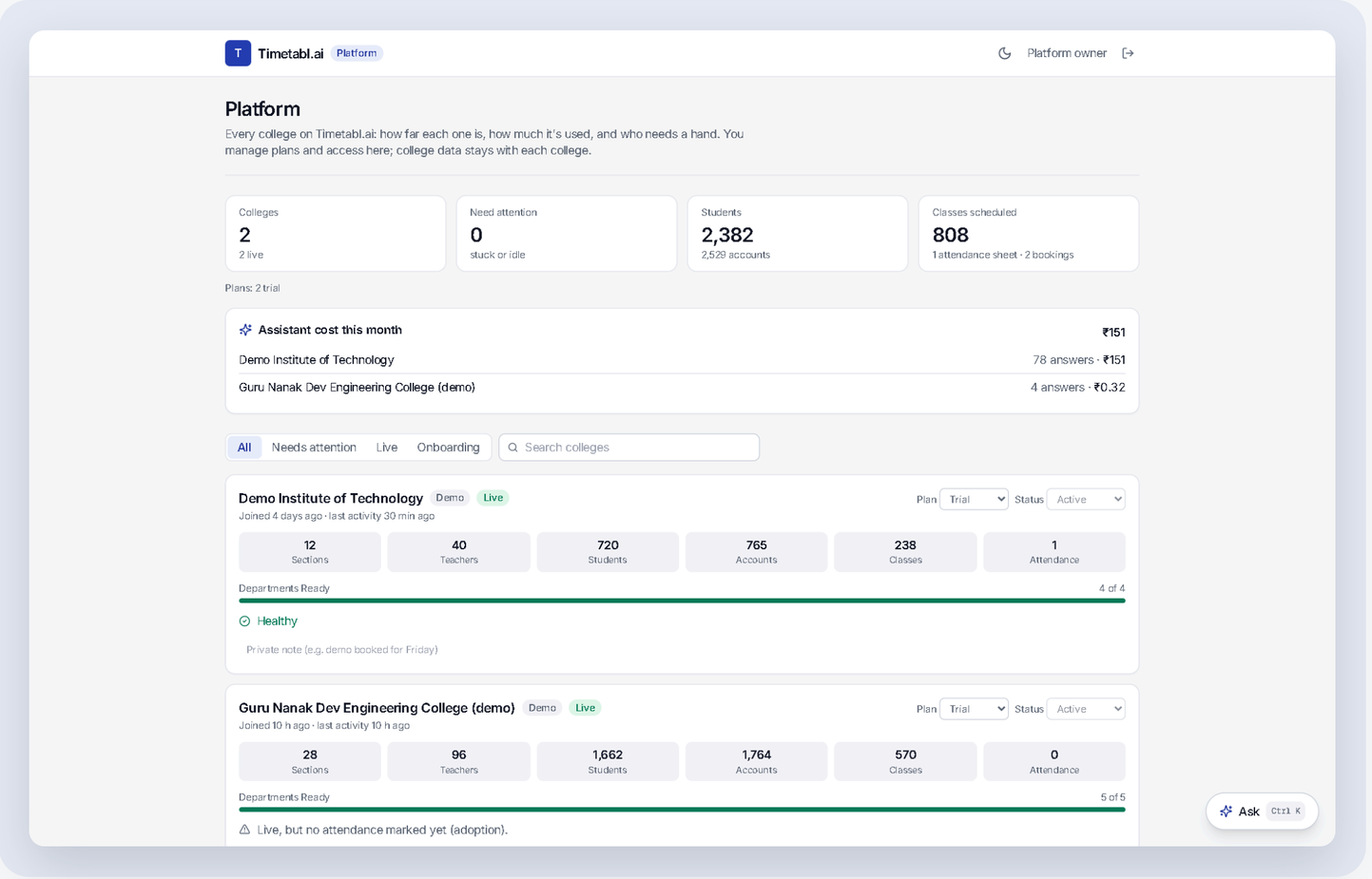
Room booking: free rooms only; the room’s Coordinator approves

READY SAAS

Many colleges, one platform

> **Each college's data stays apart. The platform owner sees every college's progress, health and AI cost, never its data.**

- **Starter:** free, one department
- **College:** ₹25 / student / year
- **University:** custom



RESULTS

Tested on two colleges, checked every time

0
clashes, on every timetable and change

238
classes · Demo Institute, 12 sections, 2 shifts

570
classes · GNDEC demo, 1,662 students

187
automated tests, all green

Independent validator: no code shared with the solver

Real edge cases: shifts, lab batches, electives, borrowed rooms

Cheap to run: instant answers free; agentic ones about ₹2

Thank you. *Try it live.*

Submission	Link
Deployed app	timetabl-ai.vercel.app · one-click demo logins
GitHub	github.com/SamiyaS5904/Hackathon-PS_1-
Testing dataset	docs/demo-data/GNDEC · password demo1234
Guides	README · Demo guide · Deploy guide

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